This full text paper was peer reviewed at the direction of IEEE Communications Society subject matter experts for publication in the IEEE ICC 2010 proceedings

All-Terminal Network Reliability Optimization in Fading Environment via Cross Entropy Method

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***Abstract*—This paper presents a new algorithm that can be readily applied to solve all-terminal network reliability optimization problem of a wireless network in a fading environment. The optimization problem solved considers finding the optimal topological layout of links at which the all-terminal network reliability is maximized by controlling the nodes' transmission powers. To that end, a link probabilistic model is developed to relate fading, attenuation, interference and nodes' transmission powers to link reliability. Then, the proposed algorithm utilized this probabilistic model to control nodes' transmission power to maximize links reliabilities and hence all-terminal network reliability. The proposed algorithm is based on two major steps that use a global stochastic optimization technique, Cross Entropy (CE) to generate the optimal network topology and control nodes transmission powers such that all-terminal network reliability is maximized. An illustrative example is used to illustrate the proposed algorithm.**

I. Introduction

An important stage of the design of communication networks is to find the best layout of components to minimize cost while meeting a performance criterion, such as transmission delay, throughput or reliability [1]. This paper focuses on wireless communication network topological design problem with the objective of maximizing the *all-terminal network reliability*, i.e., the probability that every node in the network can communicate with every other node [2]–[6].

The primary design problem is to choose a set of links for a given set of nodes to maximize all-terminal reliability given constraints on the nodes' transmission powers. This design problem is NP-hard [7], and as a further complication, the calculation of all-terminal reliability is also NP-hard [8].

This problem and related versions have been studied throughout the literature with both enumerative-based methods and heuristic methods. Jan et al. [1] developed an algorithm using decomposition based on branch and bound to minimize link costs with a minimum network reliability constraint; this approach is computationally tractable for fully connected networks up to 12 nodes. However, even for fully connected networks of a relatively small size, this approach requires considerable computation time. Tabu search method was used in [2] to obtain solutions to this problem based on surrogate formulations of all-terminal reliability. Unfortunately, these methods are ineffective for implementation in real-world problems due to the requirement of high computation load needed to enumerate, save and process every states in the network.

This research was supported by NSF Engineering Research Center for Future Renewable Electric Energy Delivery and Management (FREEDM) Systems under award 0812121.

Thus, several heuristic techniques have been developed to overcome high computation requirement. A deterministic version of simulated annealing (SA) was used in [3] with exact calculation of network reliability to find the optimal design of small networks, i.e., five nodes or less. The genetic algorithms (GAs) [4] [5] provide a good solution with low variability of the solution quality for relatively small networks, i.e., less than 10 nodes. Unfortunately, GA method has been proved to yield poor-quality solutions for networks above 10 nodes [6]. Furthermore, these methods require the development, coding, and testing of a problem-specific GA, complicating the solution process. AboElFotoh and AlSumait [6] presented a new approach based on Artificial Neural Networks (ANN) for solving the problem. The problem is mapped onto an optimization ANN by constructing an energy function whose minimization process drives the neural network into one of its stable states. This stable state corresponds to a solution for the network design problem. Although solutions are attained in a very small time, the development of ANN solution entitles an extensive time and significant parameters tuning.

This work differs from aforementioned works in various aspects. Links reliabilities are not fixed in wireless environment because they are prone to change due to fading, attenuation and interference. Thus, we utilize the signal-to-interference ratio (SIR) to relate channel fading, attenuation effect, interference and transmission power to the links reliabilities [9]–[11]. By controlling nodes' transmission powers, we control SIR to maximize links reliabilities and hence maximize all-terminal network reliability. In this paper, a node's transmission power is selected from finite set of power levels. The selection of nodes' transmission powers with the objective of maximum all-terminal reliability is variation of NP-hard optimization problem known as the Minimum Steiner Problem, which is a 0-1 knapsack problem with a non-linear objective [7]. To our knowledge, no simple algorithm is known that can tackle at the same time these three NP-hard problems: topology optimization, all-terminal network reliability calculation and all-terminal network reliability maximization. Therefore, we propose a new effective algorithm to handle this design problem. This algorithm is based on Cross Entropy (CE) [12] which is an adaptive technique for estimating rare events in complex statistics networks.

This paper is organized as follows. Section II presents the preliminaries and the problem formulation. Section III describes the details of the proposed algorithm. Section IV provides results for an illustrative example. Finally, Section V concludes the paper.

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II. Preliminaries

A. Rayleigh/Rayleigh Fading Environment

Consider a wireless network consisting of n nodes. Each node can act as a transmitter and a receiver at same time utilizing different frequency channels, codes, or antenna beams in an antenna array. The power received at node j from i is:

$$G_{ij} F_{ij} P_i^{tx}, \tag{1}$$

where G_{ij} is the path gain from node i to node j, F_{ij} is the Rayleigh fading and P_i^{tx} is node's i transmission power. F_{ij}s are assumed to be independent exponentially random variables with unit mean. Thus, the received power at node j from node i is an exponentially distributed random variable with a mean of $E[G_{ij}F_{ij}P_i^{tx}] = G_{ij}P_i^{tx}$ [9]. In this study, both desired and interference signals are subject to Rayleigh fading, i.e., receiving node has no direct line-of-sight signal component, either from the transmitting or the interfering nodes. This Rayleigh/Rayleigh fading environment is very realistic in dense urban areas [9]–[11].

B. Link Reliability in Rayleigh/Rayleigh Fading Environment

In Rayleigh/Rayleigh fading environment, SIR is:

$$SIR_j = \frac{G_{ij}F_{ij}P_i^{tx}}{\sum_{k\neq i} G_{kj}F_{kj}P_k^{tx}}, \tag{2}$$

where $G_{ij}F_{ij}P_i^{tx}$ represents the received power at node j from node i and $\sum_{k\neq i} G_{kj}F_{kj}P_k^{tx}$ is the total interference received at node j. For a receiving threshold SIR^{th}, the probability of successful communication from node i to node j is

$$\begin{aligned} Pr_{ij} =& 1 - Pr(SIR_j \leq SIR^{th}) \\ =& 1 - Pr\Big(G_{ij}F_{ij}P_i^{tx} \leq SIR^{th} \sum_{k\neq i} G_{kj}F_{kj}P_k^{tx}\Big), \end{aligned} \tag{3}$$

where $Pr\Big(G_{ij}F_{ij}P_i^{tx} \leq SIR^{th} \sum_{k\neq i} G_{kj}F_{kj}P_k^{tx}\Big)$ can be interpreted as the probability that the communication from node i to node j experiences a failure due fading. This probability is the *link outage probability* [9]–[11].

Result 1: For a wireless network with n nodes operating in a Rayleigh/Rayleigh fading environment, it was proved in [10] that the probability of successful communication from node i to j is

$$Pr_{ij} = \prod_{k\neq i} \frac{1}{1 + \frac{SIR^{th} G_{kj} P_k^{tx}}{G_{ij} P_i^{tx}}}. \tag{4}$$

C. Topology Optimization and All-Terminal Reliability

Consider a directed $G(V, E)$ graph with set V of nodes and set E of links. Suppose the number of nodes is $|V| = n$. For a fully connected network, the number of links is $|E| = n^2 - n$. A vector $\boldsymbol{e} = (e_{12}, \ldots, e_{ij}, \ldots, e_{nn-1})$ denote the directed links between nodes. The existence of e_{ij} implies the existence e_{ji} to ensure bidirectional communication between any pair of nodes. A vector $\boldsymbol{P^{tx}} = (P_1^{tx}, \ldots, P_n^{tx})$ represents the transmission powers associated with each node. The selection of P_i^{tx} and P_j^{tx} determines the probabilities of successful communication, Pr_{ij} and Pr_{ji} , between nodes i and j. In practice, wireless node may not have the capability of controlling transmission power continuously, but rather support only finite set of power levels. Suppose that a wireless node can support a finite set of powers levels, i.e., $\{\triangle P^{tx}, 2\triangle P^{tx}, \ldots, L\triangle P^{tx}\}$, where L is the number of power levels and $\triangle P^{tx} = P_{max}^{tx}/L$, where P_{max}^{tx} denote the node's maximum transmission power.

We identify links selection using link state vector $\boldsymbol{y} = (y_{12}, \ldots, y_{ij}, \ldots, y_{nn-1})$. If a link e_{ij} is selected, then its state $y_{ij} = 1$, otherwise $y_{ij} = 0$. Each state vector represents a network topology. Thus, the set of all possible state vectors represents the set of all possible network topologies. The set of of possible state vectors is denoted as Υ.

For a given state vector $\boldsymbol{y}$, let $\Phi(\cdot)$ be an indicator function that map state vector into network state as follows:

$$\Phi(y) = \begin{cases} 1 & \text{if all the nodes are connected,} \\ 0 & \text{otherwise.} \end{cases} \tag{5}$$

For a given $\boldsymbol{y}$, the all-terminal network reliability is

$$Rel(\boldsymbol{y}) = \sum_{\boldsymbol{y}\in\Upsilon} \Phi(\boldsymbol{y}) \prod_{i=1}^{n} \prod_{j=1, j\neq i}^{n} (Pr_{ij})^{y_{ij}} (1 - Pr_{ij})^{1-y_{ij}}, \tag{6}$$

Then, the problem of topology optimization with the maximization of all-terminal network reliability is:

$$\begin{aligned} \max_{\boldsymbol{y}\in\Upsilon} \quad & Rel(\boldsymbol{y}), \\ \text{subject to} \quad & 0 \leq P_i^{tx} \leq P_{max}^{tx}, \quad \forall P_i^{tx} \in \boldsymbol{P^{tx}} \\ & P_i^{tx} \in \{\triangle P^{tx}, 2\triangle P^{tx}, \ldots, L\triangle P^{tx}\}. \end{aligned} \tag{7}$$

This problem entitles three NP-hard problems. The first NP-hard problem is to identity the optimal network topology that maximizes all-terminal reliability. The second NP-hard problem is to determine the optimal nodes' transmission powers that attain the maximum all-terminal reliability for the selected network topology. The exact calculation of all-terminal reliability in (6) is the third NP-hard problem.

III. Application of CE method to Network Topology and Reliability Optimization Problem

In this section, we give a brief overview of CE method and then we demonstrate in details the proposed algorithm.

A. Cross-Entropy Method (CE): an Overview

For this paper to be self-contained, in this section we present an overview of the usage of CE method in solving an optimization problem. We refer to [12] [13] for more details.

CE method is an adaptive importance sampling derived from the associated stochastic problem (ASP) for estimating the probability of a rare-event occurrence. The estimation of this probability is determined using a log-likelihood estimator govern by a parametrized probability distribution. CE method adaptively estimates the parameters of the probability distribution which produces a random variable in the neighborhood of the globally optimal solution by minimizing cross entropy. CE method has two phases:

1) Generating candidate solutions sampled from some parametrized probability distribution.

2) Updating the parameters of the probability distribution based upon the current best solutions to bias the future search to obtain samples closer to the optimal solution.

In order to maximize an objective function $S(\boldsymbol{x})$ over a finite set of χ, i.e., $\boldsymbol{x} \in \chi$, assume that $S(\boldsymbol{x})$ has a unique maximizer $\boldsymbol{x}^*$, i.e., $\gamma^* = S(\boldsymbol{x}^*)$, then

$$S(\boldsymbol{x}^*) = \gamma^* = \max_{\boldsymbol{x}\in\chi} S(\boldsymbol{x}). \tag{8}$$

The first step in applying CE method is to associate an estimation problem with the optimization problem in (8). Thus, a collection of functions $\{H(\cdot;\gamma)\}$ on χ as follow:

$$H(\boldsymbol{x};\gamma) = \begin{cases} 1 & S(\boldsymbol{x}) \geq \gamma, \\ 0 & S(\boldsymbol{x}) < \gamma, \end{cases} \tag{9}$$

for each $\boldsymbol{x} \in \chi$ and threshold $\gamma \in \mathbb{R}$. Next, let $\{f(\cdot;\boldsymbol{v})\}$ be a family of probability mass functions (pmf's) on χ, parametrized by real-valued parameter vector $\boldsymbol{v}$. The problem in (8) is associated with the problem of estimating:

$$\begin{aligned} \ell_{\boldsymbol{v}}(\gamma) &= \mathbb{P}_{\boldsymbol{v}}(S(\boldsymbol{X}) \geq \gamma) \\ &= \sum_{\boldsymbol{x}} H(\boldsymbol{x};\gamma) f(\boldsymbol{x};\boldsymbol{v}) = \mathbb{E}_{\boldsymbol{v}} H(\boldsymbol{X};\gamma), \end{aligned} \tag{10}$$

where $\mathbb{P}_{\boldsymbol{v}}$ is a probability measure under which the random state $\boldsymbol{X}$ has probability mass function $f(\cdot;\boldsymbol{v})$, and $\mathbb{E}_{\boldsymbol{v}}$ denotes the corresponding expectation. The problem in (10) is the associated stochastic problem (ASP). The case, $\ell_{\boldsymbol{v}} = f(\boldsymbol{x^*};\boldsymbol{v})$ will be a very small number. Therefore, *Importance Sampling* technique (IS) [12] is used to estimated such small number. Using IS, a random sample $\boldsymbol{X}^{(1)},\ldots,\boldsymbol{X}^{(N)}$ is taken from a different pmf g on χ, then $\ell_{\boldsymbol{v}}(\gamma)$ is estimated as follows:

$$\ell_{\boldsymbol{v}}(\gamma) = \frac{1}{N}\sum_{k=1}^{N} H(\boldsymbol{X}^{(k)};\gamma)\frac{f(\boldsymbol{X}^{(k)};\boldsymbol{v})}{g(\boldsymbol{X}^{(k)})}. \tag{11}$$

Optimal γ^* occurs when g assigns all its probability mass to select $\boldsymbol{x^*}$. Therefore, as g assigns most of its probability mass close to $\boldsymbol{x^*}$, the obtained γ is close to γ^*. CE method will efficiently estimate $\ell_{\boldsymbol{v}}(\gamma)$ in (10) by controlling g and hence generation of state $\boldsymbol{x}$. The optimal way to estimate $\ell_{\boldsymbol{v}}(\gamma)$ is to use the change of measure with pmf :

$$g^*(\boldsymbol{x}) = \frac{H(\boldsymbol{x};\gamma)f(\boldsymbol{x};\boldsymbol{v})}{\ell_{\boldsymbol{v}}(\gamma)}. \tag{12}$$

Note that g^* in (12) depends on unknown parameter $\ell_{\boldsymbol{v}}$. To overcome this problem, an optimal pmf $f(\cdot,\boldsymbol{v^*})$ is chosen such that the distance between this pmf and g^* is minimal. The distance between two pmf's g and h is measured using *Kullback-Leibler distance*, i.e., cross-entropy [14] which is

$$\mathcal{K}(g,h) = \mathbb{E}_g \log\frac{g(\boldsymbol{X})}{h(\boldsymbol{X})} = \sum_{\boldsymbol{x}} g(\boldsymbol{x})\log g(\boldsymbol{x}) - \sum_{\boldsymbol{x}} g(\boldsymbol{x})\log h(\boldsymbol{x}). \tag{13}$$

For estimating $\ell_{\boldsymbol{v^*}}(\gamma)$ in (10), the parameter v^* is chosen such that $\mathcal{K}(g*, f(\cdot,\boldsymbol{v^*}))$, with g^* in (12), is minimal. Thus, v^* should be selected such that (14) is maximal.

$$\mathbb{E}_{\boldsymbol{v}} H(\boldsymbol{X};\gamma)\log f(\boldsymbol{X};\boldsymbol{v^*}). \tag{14}$$

In CE method, the components of $\boldsymbol{v}$ can be calculated analytically in order to drive change of measure g toward g^* to achieve γ^*. For discrete random vectors $\boldsymbol{X}$, $\boldsymbol{v}$ is

$$\boldsymbol{v} = \frac{\mathbb{E}_{\boldsymbol{v}} H(\boldsymbol{X};\gamma) I_{\{\boldsymbol{X}\in A\}}}{\mathbb{E}_{\boldsymbol{v}} H(\boldsymbol{X};\gamma) I_{\{\boldsymbol{X}\in B\}}}, \tag{15}$$

where $I_{\{\boldsymbol{X}\in A\}}$ and $I_{\{\boldsymbol{X}\in B\}}$ are indicator random variables and $A \subset B \subset \chi$. The vector $\boldsymbol{v}$ in (15) is estimated as follows

$$\boldsymbol{v} = \frac{\sum_{k=1}^{N} \mathbb{E}_{\boldsymbol{v}} H(\boldsymbol{X}^{(k)};\gamma) I_{\{\boldsymbol{X}^{(k)}\in A\}}}{\sum_{k=1}^{N} \mathbb{E}_{\boldsymbol{v}} H(\boldsymbol{X}^{(k)};\gamma) I_{\{\boldsymbol{X}^{(k)}\in B\}}}, \tag{16}$$

where $\boldsymbol{X}^{(1)},\ldots,\boldsymbol{X}^{(N)}$ is a random samples from the pmf $f(\cdot;\boldsymbol{v})$. It is important to that estimator in (16) implies implicitly that the numerator and denominator must be non-zero in order to attain optimal estimation of $\boldsymbol{v^*}$.

Now CE method will construct a sequence of parameters $\boldsymbol{v_0},\boldsymbol{v_1},\ldots$ and performance measures $\gamma_0,\gamma_1,\ldots$ such that $\boldsymbol{v_k}$ converges to a parameter vector whose corresponding pmf assigns high probability mass to the collection of states that enforces γ_k to converges to the optimal γ^*.

B. Main Algorithm

The proposed algorithm consists of two main CE procedures which are repeated until optimal topology is attained:

- CE1–Topology Optimization:
  1. Generate random state vectors $\boldsymbol{Y^1},\ldots,\boldsymbol{Y^M}$ according to parametrized random distribution.
  2. Update the parameters of the probability distribution based upon the current best solutions calculated using CE2 to bias the future generation of $\boldsymbol{Y^i}$ toward the optimal $\boldsymbol{Y^*}$.
- CE2–Maximizing All-Terminal Network Reliability:
  1. Generate random nodes' transmission powers vectors $\boldsymbol{P^{tx1}},\ldots,\boldsymbol{P^{txN}}$ according to parametrized random distribution.
  2. Update the parameters of the probability distribution based upon the current best solutions that attain the maximum all-terminal reliability for the given $\boldsymbol{Y}$ to bias the future generation of $\boldsymbol{P^{tx i}}$ toward the optimal $\boldsymbol{P^{tx*}}$.

Next, we illustrate in details the operation and interaction of the two CE procedures in solving the problem in (7).

C. CE1–Topology Optimization

An efficient method to generate random topology, i.e., state vector, $\boldsymbol{y} = (y_{12},\ldots,y_{ij},\ldots,y_{nn-1})$ which satisfy all-terminal reliability condition in (5) is as follows. First, generate a "uniform" permutation $\boldsymbol{u} = (u_{12},\ldots,u_{ij},\ldots,u_{nn-1})$ of $(1,\ldots,n^2-n)$, by s-independently drawing numbers from the uniform distribution on [0,1], and letting $(u_{12},\ldots,u_{ij},\ldots,u_{nn-1})$ correspond to the ordered observations. Second, given such a permutation, draw a Bernoulli random variable with a success probability of $u_{a_{12}}$ to decide whether to select link y_{12} or not. If successful, $y_{12} = 1$ and y_{21} is also selected to ensure bidirectional communication between nodes 1 and 2. Otherwise, y_{12}=$y_{21} = 0$. We repeat the above procedure sequentially for all links. After all links have been processed, we check whether if the state vector, $\boldsymbol{y}$ contains a spanning tree of the network using Breadth-First Search (BFS), and if so, $\boldsymbol{y}$ is accepted to be used in the calculation of the all-terminal reliability in **Algorithm 2**. The main algorithm for generating random state vectors is summarized as follows:

The main procedure for topology optimization, **Algorithm 2**, is to construct a sequence probability selection vector $\boldsymbol{u_t}$, such that $\boldsymbol{u_t}$ converges to a binary probability vector $\boldsymbol{u}^*$ that corresponds to the optimal state vector $\boldsymbol{y}^* = \boldsymbol{u}^*$. At each iteration t, for a given $\boldsymbol{u_{t-1}}$, $\gamma_t = S_{[\rho_1 M]}$ is the highest achieved reliability is obtained from the descending ordered, with respect to reliability, $\rho_1 M$ state vectors where $\rho_1 \in [0.01, 0.1]$. Using $\rho_1 M$ state vectors, the j^{th} element of $\boldsymbol{u_{t+1}}$ is obtained by calculating the ratio of number of times the j^{th} element of state vector $\boldsymbol{y}$ was selected to the total number state vector samples at which the all-terminal reliability exceed γ_t as shown in (17). Starting with $\boldsymbol{u_0} = \frac{1}{2}$, i.e., all links have equal selection probabilities, **Algorithm 2** updates $\boldsymbol{u}$ to converge to $\boldsymbol{u}^*$, such that $\boldsymbol{y}$ converges to $\boldsymbol{y}^*$, i.e., optimal network topology. CE1 procedure is terminated if for some integer d and $t \geq d$, $\gamma_t^* = \ldots = \gamma_{t-d}^*$. M is the sample size, e.g., $M = \tau n$ where $\tau > 0$. **Algorithm 2** is summarized as follows:

Algorithm 1 Random Topology Generator

1. Generate a uniform random permutation $(a_{12}, \ldots, a_{ij}, \ldots, a_{nn-1})$.
2. For all y_{ij} Draw random $y_{ij} \sim Ber(u_{a_{ij}})$. If $y_{ij} = 1$ then $y_{ji} = 1$ else $y_{ij} = y_{ji} = 0$.
3. Check If $BSF(\boldsymbol{y}) = 1$, accept the $\boldsymbol{y}$ and exit. Otherwise, go to step 1.

Algorithm 2 Main CE Algorithm for Topology Optimization

START with some $\boldsymbol{u_0} = \frac{1}{2}$. Let $t = 1$.

REPEAT

1. Draw random sample of state vectors $\boldsymbol{Y}^{(1)}, \ldots, \boldsymbol{Y}^{(M)}$ according to **Algorithm 1**, with $\boldsymbol{u} = \boldsymbol{u_{t-1}}$.
2. Calculate the all-terminal network reliability $S(\boldsymbol{y}^{(i)})$ for all i using **Algorithm 4** and sort them in descending order, $S_1 \geq \ldots \geq S_M$. Let $[\rho_1 M]$ be the integer part of $\rho_1 M$ where $\rho_1 \in [0.01, 0.1]$. Define $\gamma_t = S_{[\rho_1 M]}$.
3. Using the same sample, calculate $\boldsymbol{u}_{t+1} = (u_{t+1,j})$

$$u_{t+1,j} = \frac{\sum_{i=1}^{M} I_{\{S(\boldsymbol{Y}^{(i)}) \geq \gamma_t\}} Y_{ij}}{\sum_{i=1}^{M} I_{\{S(\boldsymbol{Y}^{(i)}) \geq \gamma_t\}}}. \tag{17}$$

Increase t by 1.

UNTIL For some $t \geq d$, $\gamma_t^* = \gamma_{t-1}^* = \ldots = \gamma_{t-d}^*$.

D. CE2–Maximizing All-terminal Network Reliability

In order to apply CE to maximize the all-terminal reliability, we need to specify how to generate random power assignments and how to update the parameters at each iteration. Let each possible transmission power assignment be denoted by a vector $\boldsymbol{x} = (x_0, \ldots, x_{n-1})$ in the set $\boldsymbol{X} = \Big\{\boldsymbol{x} = (x_0, \ldots, x_{n-1}) : x_i \in \{0, \triangle P^{tx}, \ldots, L \triangle P^{tx}\}\Big\}$. Define a function S on $\boldsymbol{X}$ such that $S(\boldsymbol{x})$ denote the all-terminal network reliability. Then, the optimal power assignment is obtained by solving:

$$\text{maximize} \quad S(\boldsymbol{x}) \; over \; \boldsymbol{x} \in \boldsymbol{X}. \tag{18}$$

CE2 procedure requires a random mechanism to generate the power level assignment for nodes. A simple method to generate a random power assignment, i.e., $\boldsymbol{x} = (x_0, \ldots, x_{n-1})$ in X, is to first draw a $x_0, \ldots, x_{n-1}$ independently, each x_i according to a $(L+1)$-point discrete distribution $(p_{i,0}, \ldots, p_{i,L}), i = 1, \ldots, n$. Note that p_{ij} represents the probability node i is assigned a transmission level power of $j \triangle P^{tx}$. Thus, a matrix $\boldsymbol{P} := (p_{ij})$ with dimension of $n \times L+1$ determine which power level should be assigned to which node. Note that the row of $\boldsymbol{P}$ sum up to one. The pmf $f(\cdot; \boldsymbol{P})$ of $\boldsymbol{X}$ is thus parametrized by matrix $\boldsymbol{P}$ and given by

$$f(\boldsymbol{x}; \boldsymbol{P}) = \prod_{i=0}^{n-1} \sum_{j=0}^{L} p_{ij} 1_{\boldsymbol{X} \in \boldsymbol{X_{ij}}}, \tag{19}$$

where $\boldsymbol{X_{ij}} = \{\boldsymbol{X_{ij}} \in \boldsymbol{X} : x_i = j \triangle P^{tx}\}$, i.e., $\boldsymbol{X}_{ij}$ is the i-th coordinate of $\boldsymbol{x}$ is $j \triangle P^{tx}$.

Starting with some initial value matrix $\boldsymbol{P_0}$, CE2 procedure updates elements $\boldsymbol{P} := (p_{ij})$ to converges to $\boldsymbol{P}^*$, such that each row, i, of $\boldsymbol{P}^*$ contains only one element, j, equals to one and the remaining are zeros, i.e., the power assignment $j \triangle P^{tx}$ to node i. The non-zero elements define the power assignment which maximizes the all-terminal network reliability. The updating rule for $\boldsymbol{P}$ is derived from the maximization of (14) with the condition that the sum of each row is one. Applying Lagrangian optimization to (14), the following maximization problem is obtained:

$$\max_{\boldsymbol{P}, \lambda_0, \ldots, \lambda_{n-1}} \Big[\mathbb{E}_{\boldsymbol{P}} H(\boldsymbol{X}; \gamma) \log f(\boldsymbol{X}; \boldsymbol{P}) + \sum_{i=0}^{n-1} \lambda_i \Big(\sum_{j=0}^{L} p_{ij} - 1\Big)\Big], \tag{20}$$

where $\lambda_0, \ldots, \lambda_{n-1}$ are Lagrange multipliers. Differentiating with respect to p_{ij} yields:

$$\mathbb{E}_{\boldsymbol{P}} \frac{H(\boldsymbol{X}; \gamma) I_{\{\boldsymbol{X} \in \boldsymbol{X_{ij}}\}}}{p_{ij}} + \lambda_i = 0, \quad \forall j = 0, \ldots, L. \tag{21}$$

Then, summing over all j yields $\mathbb{E}_{\boldsymbol{P}} H(\boldsymbol{X}; \gamma) = -\lambda_i$ so that

$$p_{ij} = \frac{\mathbb{E}_{\boldsymbol{P}} H(\boldsymbol{X}; \gamma) I_{\{\boldsymbol{X} \in \boldsymbol{X_{ij}}\}}}{\mathbb{E}_{\boldsymbol{P}} H(\boldsymbol{X}; \gamma)} \tag{22}$$

For N samples, we can estimate (22) as follows:

$$p_{ij} = \frac{\sum_{k=1}^{N} I_{\{S(\boldsymbol{x}^{(k)}) \geq \gamma\}} I_{\{\boldsymbol{X}^{(k)} \in \boldsymbol{X_{ij}}\}}}{\sum_{k=1}^{N} I_{\{S(\boldsymbol{x}^{(k)}) \geq \gamma\}}}. \tag{23}$$

This estimator is the ratio of the number of times $\boldsymbol{x}^{(i)}$ obtain value greater than γ_k, to the number of times their i-th coordinate equal to j. **Algorithm 3** generates power level at each node using (23). The remaining unspecified parameters of the CE2 procedure are initial matrix $\boldsymbol{P_0}$, the stopping criterion and N. For $\boldsymbol{P_0}$, we set all its elements equal to $\frac{1}{L+1}$. The stopping criterion is the convergence of matrix $\boldsymbol{P}$ to $\boldsymbol{P}^*$, such that each row, i, of $\boldsymbol{P}^*$ contains only one element, j, equals to one and the remaining are zeros. The convergence of $\boldsymbol{P}$ is

Algorithm 3 Random Power Assignment Algorithm

For a given $\mathbf{P}$=p_{ij}, generate a random permutation $(\pi_0, \ldots, \pi_{n-1})$ of $\{0, \ldots, n-1\}$.
for i = 0 to $n-1$ **do**
  $t = \sum_{j=0}^{L} P_{\pi_i,j}$
  for j = 0 to L **do**
    $P_{\pi_i,j} = P_{\pi_i,j}/t$
  end for
  Generate X_{π_i} according to $(p_{\pi_i,0}, \ldots, p_{\pi_i,L})$
end for

associated with the convergence of the all-terminal reliability, i.e., γ^*, to a certain value. Thus, CE2 procedure is terminated if for some integer d and $k \geq d$, $\gamma_k^* = \gamma_{k-1}^* = \ldots = \gamma_{k-d}^*$. N is the sample size which can be derived from $\boldsymbol{P}$'s dimensions, e.g., $N = \tau n(L+1)$ where τ is an integer and $\tau > 0$.

Algorithm 4 Main CE Algorithm for Reliability Optimization

START with some $\boldsymbol{P_0} = \frac{1}{L+1}$. Let $k = 1$.
REPEAT

1: Draw a random sample of power assignments $\boldsymbol{X}^{(1)}, \ldots, \boldsymbol{X}^{(N)}$ according to **Algorithm 2**, with $\boldsymbol{P} = \boldsymbol{P_{k-1}}$.
2: Calculate the all-terminal network reliability $S(\boldsymbol{X}^{(i)})$ for all i, and sort them in descending order, $S_1 \geq \ldots \geq S_N$. Let $[\rho_2 N]$ be the integer part of $\rho_2 N$ where $\rho_2 \in [0.01, 0.1]$. Define $\gamma_k = S_{[\rho_2 N]}$.
3: Using the same sample, calculate $\boldsymbol{P}_{k+1} = (P_{k+1,ij})$

$$P_{k+1,ij} = \frac{\sum_{i=1}^{N} I_{\{S(\boldsymbol{x}^{(i)}) \geq \gamma_k\}} I_{\{\boldsymbol{X^{(i)}} \in \boldsymbol{X_{ij}}\}}}{\sum_{i=1}^{N} I_{\{S(\boldsymbol{x}^{(i)}) \geq \gamma_k\}}}. \tag{24}$$

Increase k by 1.

UNTIL For some $k \geq d$, $\gamma_k^* = \gamma_{k-1}^* = \ldots = \gamma_{k-d}^*$.

IV. Illustrative Example

Consider a 5-node network which has been studied in [6] in maximizing all-terminal reliability subject to links costs. We utilize the links cost as distance metric between nodes. Thus, vector $|\boldsymbol{e}| = (32, 54, 62, 25, 34, 58, 45, 36, 52, 29)$ corresponds to the length of links ($e_{12}, e_{13}, e_{14}, e_{15}, e_{23}, e_{24}, e_{25}, e_{34}, e_{35}, e_{45}$). Note that $|e_{ij}| = |e_{ji}|$.

We used the following network parameters, $SIR^{th} = 0$ dBm, $P_{max}^{tx} = 20$ dBm and $L = 20$, where $\triangle P^{tx} = 1$ dBm. For topology optimization, CE1 parameters are $M = 10n = 50$, $\rho_1 = 0.1$, and we took $\boldsymbol{u} = \frac{1}{2}$. For all-terminal reliability optimization, CE2 parameters are $N = 20nL = 2000$, $\rho_2 = 0.1$, and, we took $\boldsymbol{P_0} = (\frac{1}{21})\ \forall p_{ij} \in \boldsymbol{P_0}$. For this study, the GA parameters are as follows: the initial population is 1024, the selection rate is 0.5 and mutation rate is 0.20.

Table IV show how our algorithm rapidly finds the optimal network topology at the third iteration, and also achieves the most appropriate nodes' power transmission assignment that maximizes both links and network reliabilities.

CE method outperformed Simulating Annealing (SA) and Genetic Algorithm (GA) in finding the optimal node power assignment per candidate topology as shown in Table IV. Our algorithm is twice faster then GA and searches only 0.31% of total search space for a given topology. Simple SA has shown the worst performance because it only uses the geometric decay method in predicating the next optimal solution which leads the SA to be trapped in sub-optimal solutions instead of approaching the optimal one [3].

TABLE I
Illustrative example

	Node Power (dBm)						
	v_1	v_2	v_3	v_4	v_5	Rel(y)	$\boldsymbol{y}$
1	17	11	7	16	2	0.8699614	0010000110110000100100010
2	14	7	15	15	5	0.8699601	0010100100110000000110010
3	15	9	12	19	2	0.8699613	0010100110110000100010000
4	17	11	7	14	2	0.8699614	0010000110110000100100010
5	17	12	8	15	3	0.8699614	0010000110110000100100010

TABLE II
Comparison of Search efforts

Total search space/topology	Average solutions searched/topology		
$(L+1)^n$	CE1/CE2	GA	SA
4.0841E06	12549	25269	103084

V. Conclusion

We proposed a new algorithm based on CE method to solve all-terminal reliability optimization problem of a wireless network in a fading environment. We explained in details how CE procedures are interacting in controlling nodes transmission power to find optimal topology, maximize all-terminal reliability and outperform SA and GA algorithms.

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